THE UK UNIVERSITY INTEGRATION BEE 2024/2025

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ROUND TWO GROUP ROUND

Sponsored by



Jane Street

1. $\displaystyle\int_0^1 \frac{\mathrm{d}x}{1+(x+x^2+x^3+x^4+x^5+...)^{2024}}$
2. $\displaystyle\int \frac{\ln(e^x-1)-x}{e^x-1}\,\mathrm{d}x$
3. $\displaystyle\int_{-\infty}^{\infty} \frac{\cos(x)+x\sin(x)}{x^2+\cos^2(x)}\,\mathrm{d}x$
4. $\displaystyle\int_0^{\frac{\pi}{2}} \frac{\arctan(\sin(x))}{\sin(x)}\mathrm{d}x$
5. $\displaystyle\int_0^1 \ln(\sqrt{1+\sqrt{x}}+\sqrt{1-\sqrt{x}})\mathrm{d}x$
6. $\displaystyle\int_0^e f(e^x)\,\mathrm{d}x$

   where $f(x)$ is the unique value $c \geq 1$ such that $c^c = x$.
7. $\displaystyle\int_0^{\infty} \frac{\sin(x)\sin(\sin(x))}{x^2}e^{\cos(x)}\,\mathrm{d}x$
8. $\displaystyle\int_0^1 \operatorname*{argmax}_{y\in\mathbb{R}} \left(x^{\sqrt{1-x^4}y^2+\sqrt{1+x^2}y-\sqrt{1-x^2}y-\sqrt{1-x^8}}\right)\mathrm{d}x$

   where $\displaystyle\operatorname*{argmax}_{y\in A}(f(y))$ is the value of $y \in A$ that maximises f.
9. $\displaystyle\int_0^{\infty} \frac{\sin x}{e^x-1}\,\mathrm{d}x$
10. $\displaystyle\int_0^1 \frac{1}{1+x}\log\left(\frac{1+3x}{2-x}\right)\,\mathrm{d}x$